STAT214 Chapter 4 Data Preparation

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1 Introduction to Data Preparation

1.1 Data Cleaning

- **Fix data issue**: fix unclear column names, extract the relevant portions of text entries, and convert the dataset to a format that is easier to work with
- **Making judgment calls** is unavoidable
- **Involve obtaining a detailed understanding** of your data and its limitations

1.2 Preprocessing

- The process of further modifying your clean data to **fit a particular algorithm or analysis**
- involve imputing missing values, converting categorical variables to numeric formats, or featurization

1.3 Overall Process of Data Preparation

1. **cleaning** your data by modifying it so it is appropriately formatted and unambiguous
2. **preprocessing** your data so it conforms to the formatting requirements required by whatever computational algorithms you plan to use to answer your question

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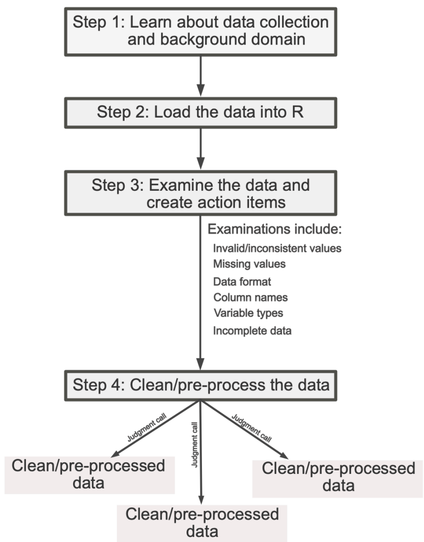
2 A Generalizable Data Cleaning Procedure

2.1 Action Items

Action items are customized modifications that you will need to apply to your data to clean it

2.2 Four-step Data Cleaning Procedure

- Step 1: Learn about the data collection process and the problem domain
- Step 2: Load the data
- Step 3: Examine the data and create action items
- Step 4: Clean the data



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3 Step 1: Learn About the Data Collection Process and the Problem Domain

3.1 Learning About the Domain Area

- What does each variable measure?
- How was the data collected?
- What are the observational units?
- Is the data relevant to my project?
- What questions do I have, and what assumptions am I making?

4 Step 2: Load the Data

4.1 Identify How to Load Data

- How your data is being stored?
- If your data contains multiple tables, is there a key variable that connects them?
- What parts of the data are relevant to the question being asked?

4.2 Checking That Data Has Been Loaded Correctly

- Look at the first and last 10 rows of your loaded data and compare it to the first 10 rows of the raw data file to confirm that they match
- Check that the dimension of your data matches what you expect

5 Step 3: Examine the Data and Create Action Items

5.1 Common Symptoms of Messy Data

- Invalid or inconsistent values
- Improperly formatted missing values
- Nonstandard data format
- Messy column names
- Improper variable types
- Incomplete data

5.2 Messy Data Trait 1: Invalid or Inconsistent Values

5.2.1 Suggested Examination

- Look at randomly selected rows in the data, as well as sequential rows
- Print out the **minimum, maximum, and average values** of each numeric column
- Look at **histograms** of the distributions of each numeric variable
- Print out the **unique values** of categorical variables

5.2.2 Possible Action Items

- Leave them as they are
- Replace the invalid/inconsistent values with the “correct” value
- Replace invalid values with missing values
- Convert measurements with different units to a common unit

5.3 Messy Data Trait 2: Improperly Formatted Missing Values

5.3.1 Suggested Examination

- Print out the **minimum and maximum** of each numeric column
- Print out the **unique values** of categorical variables
- Look at **histograms** of the distributions of each numeric variable
- Print the **number (and the proportion) of missing values** in each column
- **Visualize the patterns of missing data** (e.g., using a heatmap)

5.3.2 Possible Action Items

- Replace ambiguous missing values (e.g., `99999` or `"missing"`) with explicit missing values (`NA`)
- Replace them with "right" values and justify your choice

⚠ Remark: A Note on Imputing Missing Values

Unless it is absolutely necessary (or your data contains a lot of observations—at least in the tens of thousands), we don't recommend imputing variables that have more than half of their values missing.

5.4 Messy Data Trait 3: Nonstandard Data Format

5.4.1 Suggested Examination

- What are the observational units? Is there a single row for each observational unit?
- Does each column correspond to a single type of measurement?

5.4.2 Possible Action Items

- Pivot your data to a wider format if it is in a long format, or to a longer format if it is in an overly wide format
- Aggregate multiple measurements for individual observational units
- Separate columns that contain merged measurements

5.5 Messy Data Trait 4: Messy Column Names

5.5.1 Suggested Examination

Simply print the column names

5.5.2 Possible Action Items

Modify column names to follow the style guide:

- lowercase
- human-readable
- words should be separated by underscores

5.6 Messy Data Trait 5: Improper Variable Types

5.6.1 Suggested Examination

- Print the type of each column
- Look at individual values in each column

5.6.2 Possible Action Items

- Converting each variable to the most appropriate type
- Reformatting the levels of categorical variables
- Aggregating uncommon/rare levels of a categorical variable

5.7 Messy Data Trait 6: Incomplete Data

5.7.1 Suggested Examination

- Check that every observational unit appears in the data **exactly once**
- Check that the number of rows in the data matches what you expect, such as from the data documentation

5.7.2 Possible Action Items

If any observational units are missing from the data, add them to the data and populate unknown entries with missing values

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6 Step 4: Clean the Data

6.1 Suggestion

Unless your cleaning process will be particularly computationally intensive, we do not recommend saving a copy of your clean datasets to load for your analysis

6.2 Additional Common Preprocessing Steps

- Standardizing or normalizing variables that are on different scales
- Transforming variables so they follow a more Gaussian
- Splitting categorical variables into binary variables, called “one-hot encoding” in ML
- Featurization